

Fine-Tuning or Retrieval?

Comparing Knowledge Injection in LLMs

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Abstract

Large language models (LLMs) encapsulate a vast amount of factual information within their pre-trained weights, as evidenced by their ability to answer diverse questions across different domains. However, this knowledge is inherently limited, relying heavily on the characteristics of the training data. Consequently, using external datasets to incorporate new information or refine the capabilities of LLMs on previously seen information poses a significant challenge. In this study, we compare two common approaches: unsupervised fine-tuning and retrieval-augmented generation (RAG). We evaluate both approaches on a variety of knowledge-intensive tasks across different topics. Our findings reveal that while unsupervised fine-tuning offers some improvement, RAG consistently outperforms it, both for existing knowledge encountered during training and entirely new knowledge. Moreover, we find that LLMs struggle to learn new factual information through unsupervised fine-tuning, and that exposing them to numerous variations of the same fact during training could alleviate this problem.

Keywords: LLMs, NLP, Fine-Tuning vs. RAG, Knowledge and Factuality.

1. Introduction

Large language models (LLMs) are able to capture vast amounts of factual information (Petroni et al., 2019; Cohen et al., 2023; Hu et al., 2023). LLMs exhibit a remarkable level of knowledge in various domains due to their massive pre-training datasets. However, there are two significant limitations to this knowledge. First, it is static and does not update with time. Second, it is non-specific and thus

may lack nuanced expertise in particular domains. While these are two different problems, they are deeply related since their solution is the same: enhancing the model’s knowledge.

Recently, the idea of adapting LLMs to particular domains and updating their knowledge has become increasingly common (Yu et al., 2022). Various models have been suggested to improve factual knowledge and capabilities in diverse fields such as healthcare (Singhal et al., 2023a;b; Wu et al., 2023a), finance (Wu et al., 2023b; Yang et al., 2023), and law (Huang et al., 2023; Nguyen, 2023).

In this work, we focus on the evaluation of a model’s knowledge and its ability to memorize, understand, and retrieve factual data. We aim to understand the concept of *knowledge injection* (Wang et al., 2020; Chen et al., 2022; Liu et al., 2020; Lauscher et al., 2020). Given some knowledge base in the form of a text corpus, what is the best way to teach a pre-trained model this knowledge?

One way to add knowledge to a pre-trained model is through fine-tuning. With fine-tuning, we continue the model’s training process and adapt it using task-specific data. By exposing the model to a specific knowledge base, we expect the model weights to adapt accordingly. This process is meant to optimize the model for targeted applications, enhancing its performance and contextual relevance in specialized domains.

Another method to enhance a model’s knowledge base is through the use of in-context learning (ICL) (Chen et al., 2021; Radford et al., 2019; Min et al., 2021; Lampinen et al., 2022). The main idea behind ICL is to improve the performance of pre-trained LLMs on new tasks by modifying the input query to the model without directly changing the weights of the model. One form of ICL is retrieval augmented generation (RAG) (Lewis et al., 2020; Neelakantan et al., 2022). RAG uses information retrieval techniques to enable LLMs to obtain relevant information from a knowledge source and incorporate it into generated text.

This study aims to evaluate the knowledge injection capa-

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bilities of LLMs through a comparison of fine-tuning and RAG. To illustrate the rationale, let us use an analogy. Consider three college students taking a test on a specific topic. All had access to class materials but didn't know the topic beforehand. The first student had the textbook only during the test, the second had pre-test access and studied, and the third lost access upon the test announcement. Who would probably perform better?

2. Background

To assess *knowledge injection*, we must first understand what *knowledge* means for LLMs.

Knowledge and Language Models Defining knowledge is a complex philosophical task far beyond the scope of this research. However, we can examine what factual knowledge means in the context of language models. If a model knows a fact, it can accurately and consistently answer questions about it. Furthermore, it can reliably distinguish between true and false statements related to this fact. We can then extend this definition to a whole knowledge base, not just a single fact.

Mathematically, let $\mathcal{Q} = \{q_n\}_{n=1}^N$ be a set of N multiple choice factual questions, where each question has L possible answers and exactly one correct answer. Let $\mathcal{A} = \{(a_n^1, \dots, a_n^L)\}_{n=1}^N$ be the corresponding set of possible answers, and $\mathcal{C} = \{c_n\}_{n=1}^N$ be the correct answers.

Let $\mathcal{M}$ be a language model. We denote by $\mathcal{M}(q_n) \in \{a_n^1, \dots, a_n^L\}$ the predicted answer of the model to the n -th question.

We define the *knowledge score* $\mathcal{L}$ of $\mathcal{M}$ in relation to $\mathcal{Q}$ to be the standard accuracy score:

$$\mathcal{L}_{\mathcal{M}, \mathcal{Q}} := \frac{\#\{q_n \mid \mathcal{M}(q_n) = c_n\}}{N}. \quad (1)$$

We say that the model $\mathcal{M}$ possesses *any* knowledge regarding the set of questions $\mathcal{Q}$ if the following holds:

$$\mathcal{L}_{\mathcal{M}, \mathcal{Q}} > \frac{1}{L}. \quad (2)$$

In simpler terms, the model can consistently give correct answers, outperforming a simple random guessing baseline. Naturally, if the knowledge score $\mathcal{L}_{\mathcal{M}, \mathcal{Q}}$ is higher for one model compared to another, then we assert that the former is more knowledgeable with regards to $\mathcal{Q}$ compared to the latter.

Previously Seen Knowledge One important distinction to make is between knowledge that the model has been exposed to before during pre-training as opposed to entirely new facts. Considering the size of modern LLM

training sets, they cover a vast amount of information available through web-sourced text. As a result, even in niche domains, the goal of knowledge injection is not necessarily to teach the model entirely new facts but rather to "refresh" its memory by inducing a bias toward a particular domain.

Knowledge and Reasoning We emphasize that this knowledge evaluation framework for LLMs is imperfect. Importantly, it doesn't address other quality metrics influencing a model's response. Creating a purely knowledge-intensive dataset without involving some level of reasoning is challenging. Consequently, a model with robust reasoning abilities might excel on unfamiliar knowledge-intensive tasks by making "educated guesses" in a multiple-choice exam. Therefore, any evaluation of knowledge in LLMs should consider this, with results seen as part of a broader range of benchmarks for reasoning (Sakaguchi et al., 2021), reading comprehension (Dua et al., 2019), and general language abilities (Srivastava et al., 2022). However, this evaluation framework still strongly emphasizes factual information above all else.

Causes for Factual Errors There are many possible reasons for the failure of models to answer factual questions accurately. In (Wang et al., 2023), Wang et al. introduce a taxonomy of five main model-level causes:

- **Domain knowledge deficit:** A language model may lack comprehensive expertise in a specific domain to which it has not been exposed. For example, a model trained exclusively on texts written by William Shakespeare would perform poorly when asked about the works of Mark Twain.
- **Outdated Information:** LLMs invariably have a cutoff date determined by their training dataset. Consequently, any events, discoveries, or changes occurring after the last training update will not be within the model's knowledge without access to external sources.
- **Immemorization:** Sometimes, a model is exposed to knowledge during its training process but does not retain it. This is especially true for rare facts that appear in the training dataset only scarcely (Kandpal et al., 2023).
- **Forgetting:** Language models often undergo additional training after the pre-training phase (fine-tuning). In some cases, this might lead to a phenomenon called *catastrophic forgetting* (Kirkpatrick et al., 2017; Goodfellow et al., 2013; Chen et al., 2020; Luo et al., 2023), where models lose some of the knowledge they had prior to the fine-tuning process.
- **Reasoning Failure:** In certain instances, a language model might possess relevant knowledge about a fact but fail to utilize it properly. This is particularly evident in

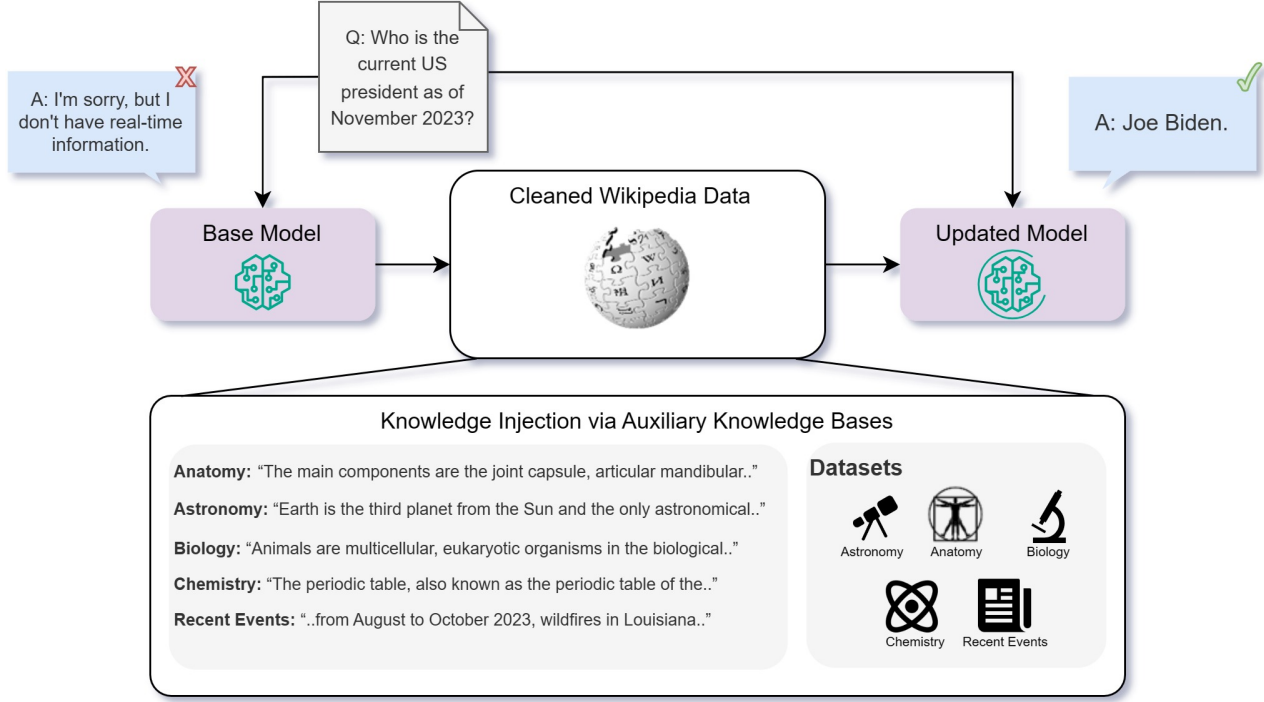


Figure 1. A visualization of the knowledge injection framework.

complex multi-step reasoning tasks (Tan et al., 2023) or when posed with different questions about the same fact, resulting in disparate outcomes (Berglund et al., 2023).

We observe that most of these issues arise during the pre-training phase, with catastrophic forgetting being the notable exception. Hence, many LLMs will suffer from factual errors of this kind regardless of any post-training process.

3. Injecting Knowledge to Language Models

Following the background given in Section 2, it is clear that general pre-training is insufficient for many knowledge-intensive tasks. To solve this, an additional post-processing step is essential to augment the knowledge of a pre-trained model. This step is often referred to as *knowledge injection* (Wang et al., 2020; Chen et al., 2022; Liu et al., 2020; Lauscher et al., 2020).

In this section, we examine two widely used frameworks for knowledge injection: fine-tuning (FT) and retrieval augmented generation (RAG). We begin by formulating the knowledge injection problem, aiming to explain both methods using consistent terminology.

3.1. Problem formulation

In Equations (1) and (2), we presented a formulation for knowledge in language models through the lens of question-

answering (Q&A). We now extend this formulation to the problem of knowledge injection using the same terminology.

Given a set of factual questions, there exists some text corpus containing information that is relevant to these questions. The central assumption of knowledge injection is that given full access to this corpus, it could serve as an auxiliary knowledge base and improve the model’s performance on this set of questions.

Mathematically, let $\mathcal{M}$ be a pre-trained model, and let $\mathcal{Q}$ be a set of factual questions, as before. Now, assume we have a relevant auxiliary knowledge base $\mathcal{B}_{\mathcal{Q}}$. Our objective is to discover a transformation, denoted as $\mathcal{F}$, that, when applied, would enhance the knowledge about $\mathcal{Q}$:

$$\mathcal{M}' := \mathcal{F}(\mathcal{M}, \mathcal{B}_{\mathcal{Q}}) \quad s.t. \quad \mathcal{L}_{\mathcal{M}', \mathcal{Q}} > \mathcal{L}_{\mathcal{M}, \mathcal{Q}}. \quad (3)$$

In this work, we aim to compare two choices for $\mathcal{F}$: fine-tuning and RAG to see which option performs better in this problem.

3.2. Fine-Tuning

Fine-tuning is the process of adjusting a pre-trained model on a specific, often narrower, dataset or task to enhance its performance in that particular domain. Here, it is vital to distinguish between different types of fine-tuning. FT techniques are commonly classified into supervised, unsu-